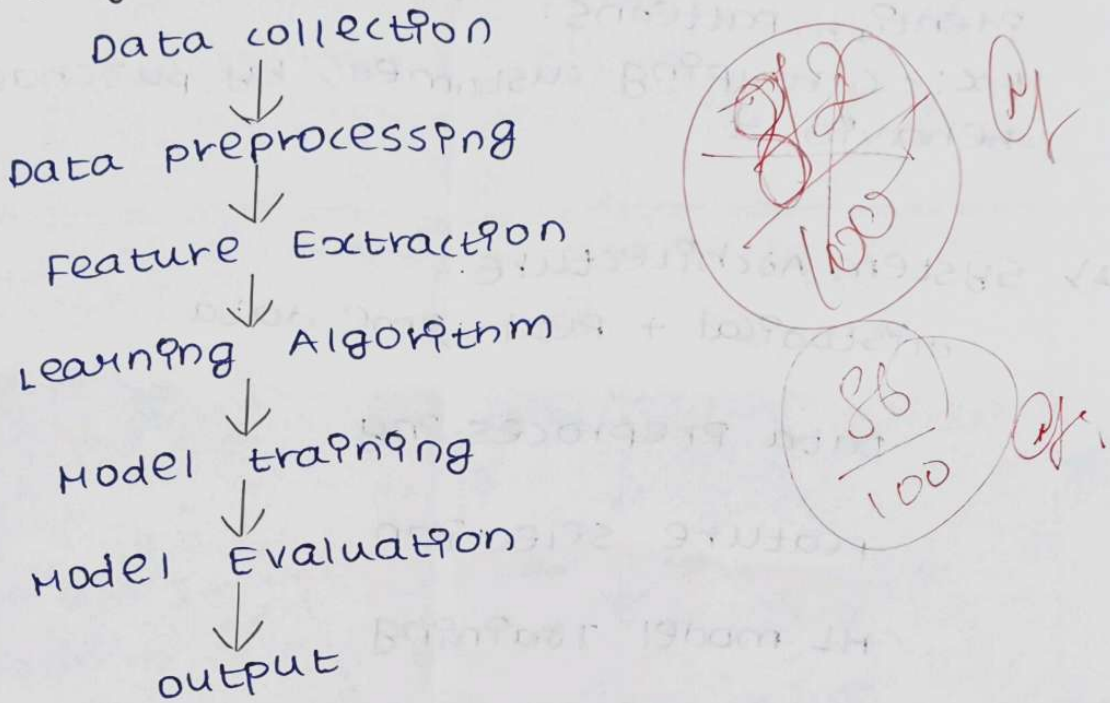
CO1-AT2
concept Mapping

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Machine Learning

1) Block Diagram:



Explanation:

- Data collection : collects data from databases, sensors or files.
- Data preprocessing : removes duplicate data and handles missing values
- Feature Extraction : Identifies important features from data.
- Learning Algorithm : Learns patterns from the dataset
- Model training : Trains the model using the prepared data.
- Model Evaluation : checks the model's performance
- prediction : Predicts output for new data.

Supervised Learning: Uses labelled data.
Ex: classifying emails as spam or not spam.

Unsupervised Learning: Uses unlabelled data to identify patterns.
Ex:- Grouping customers by purchasing behaviour.

2) System Architecture:-

Historical + Real-time data.
↓
Data Preprocessing
↓
Feature selection.
↓
ML model Training.
↓
Risk prediction.
↓
Early Warning.
↓
Emergency Response

Explanation:-

- Model selection: Random Forest classifies flood risk as low, medium or high.
- Data preprocessing: Handle missing values, remove duplicates and correct inconsistent data.
- Feature selection: select important features such as rainfall, river level, soil moisture and historical disaster records.

Early warning and Emergency response:

The system analyses real-time data and estimates disaster risk. If high risk is detected, alerts are sent through SMS, mobile applications or sirens. Authorities can plan evacuations and allocate rescue teams.

3) Step 1:

Initial Hypothesis

$h_0 = \{\emptyset, \emptyset, \emptyset, \emptyset\}$.

Step 2:

process the training examples.

Ex 1: sunny, Hot, High, weak → NO.

Ex: 2: sunny, Hot, High, strong → NO.

Ex 3: overcast, Hot, High, weak → yes.

$h_1 = \langle$overcast, Hot, High, weak$\rangle$.

Ex 4: Rainy, Mild, High, weak → yes

$h_2 = \langle ?, ?, $ High, weak $\rangle$

Ex 5: Rainy, cool, Normal, weak → yes.

$h_3 = \langle ?, ?, ?,$ weak $\rangle$

Ex 6: Rainy, cool, Normal, strong → NO.

Ex 7: overcast, Mild, Normal, strong → yes.

$h_4 = \langle ?, ?, ?, ? \rangle$

Ex 8: sunny, Mild, High, weak → NO.

Final hypothesis:

$h_f = \langle ?, ?, ?, ? \rangle$

conclusion:

The final hypothesis contains no restriction on outlook, temperature, humidity or wind because the positive examples collectively require all four attributes.

4)

S : Most specific consistent hypothesis

G : Most general consistent hypothesis

step 1:

Initialize the boundaries.

$S_0 = \{\langle \phi, \phi, \phi, \phi \rangle\}$.

$G_0 = \{\langle ?, ?, ?, ? \rangle\}$.

step 2:

process Ex 1

sunny, warm, Normal, weak → yes

$S_1 = \{\langle sunny, warm, Normal, weak \rangle\}$

$G_1 = \{\langle ?, ?, ?, ? \rangle\}$

step 3:

process Ex 2

sunny, warm, High, strong → yes

$S_2 = \{\langle sunny, warm, ?, ? \rangle\}$

$G_2 = \{\langle ?, ?, ?, ? \rangle\}$.

Step 4:
Process Ex 3.
Rainy, cold, High, Weak → NO.

$S_3 = \{<sunny, Warm, ?, ?>\}$.

$G_{13} = \{<sunny, ?, ?, ?>, <?, Warm, ?, ?>\}$.

Step 5:
Process Ex 4:
sunny, cold, Normal, Weak → yes.

$S_4 = \{<sunny, ?, ?, ?>\}$.

$G_{14} = \{<sunny, ?, ?, ?>\}$

Step 6:
Process Ex 5:
Rainy. Warm, Normal, strong → NO.

$S_5 = \{<sunny, ?, ?, ?>\}$

$G_{15} = \{<sunny, ?, ?, ?>\}$

Specific Boundary:

$S = \{<sunny, ?, ?, ?>\}$

General Boundary.

$G_1 = \{<sunny, ?, ?, ?>\}$